

Digital Forensics Case Investigation

Windows Host-Based Investigation

This case involves the forensic analysis of a compromised Windows system after the execution of a malicious attack script. The goal of this investigation is to perform host-based forensic triage, analyze key Windows Registry artifacts, identify user actions that led to the attack, reconstruct the execution chain, and determine persistence mechanisms on the system.

The investigation was conducted as a series of structured laboratory tasks, each focusing on specific artifacts and forensic techniques commonly used in incident response.

Scope of Investigation

The investigation focused on:

- Rapid triage collection using KAPE
- Registry analysis of SYSTEM, SOFTWARE, SAM, and NTUSER.DAT hives
- Identification of executed scripts and binaries
- Correlation of user activity leading to attack execution
- Detection of persistence mechanisms
- Documentation of findings and remediation actions

Tools Used

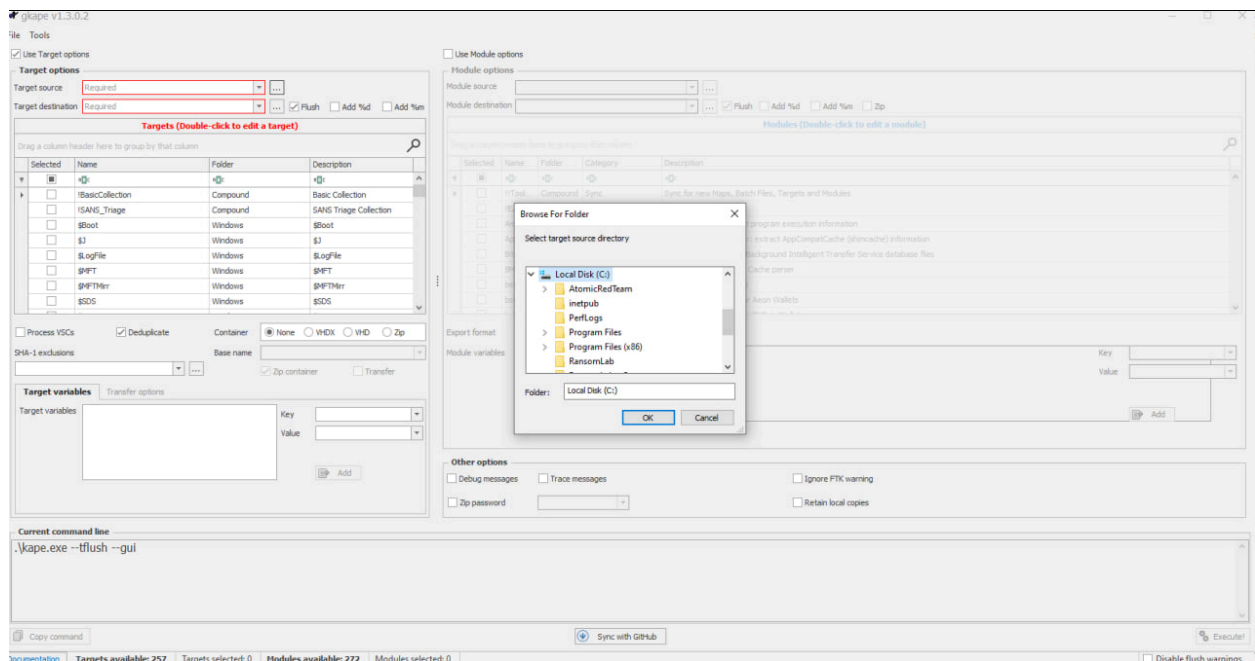
The following tools were used throughout the investigation:

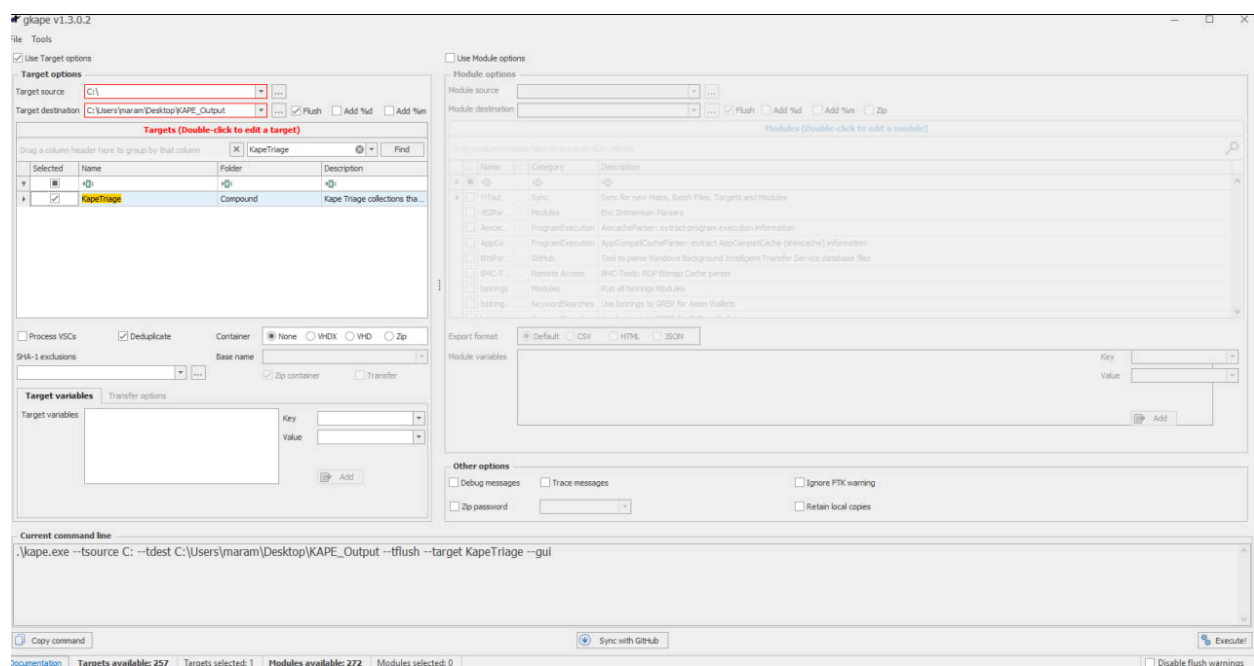
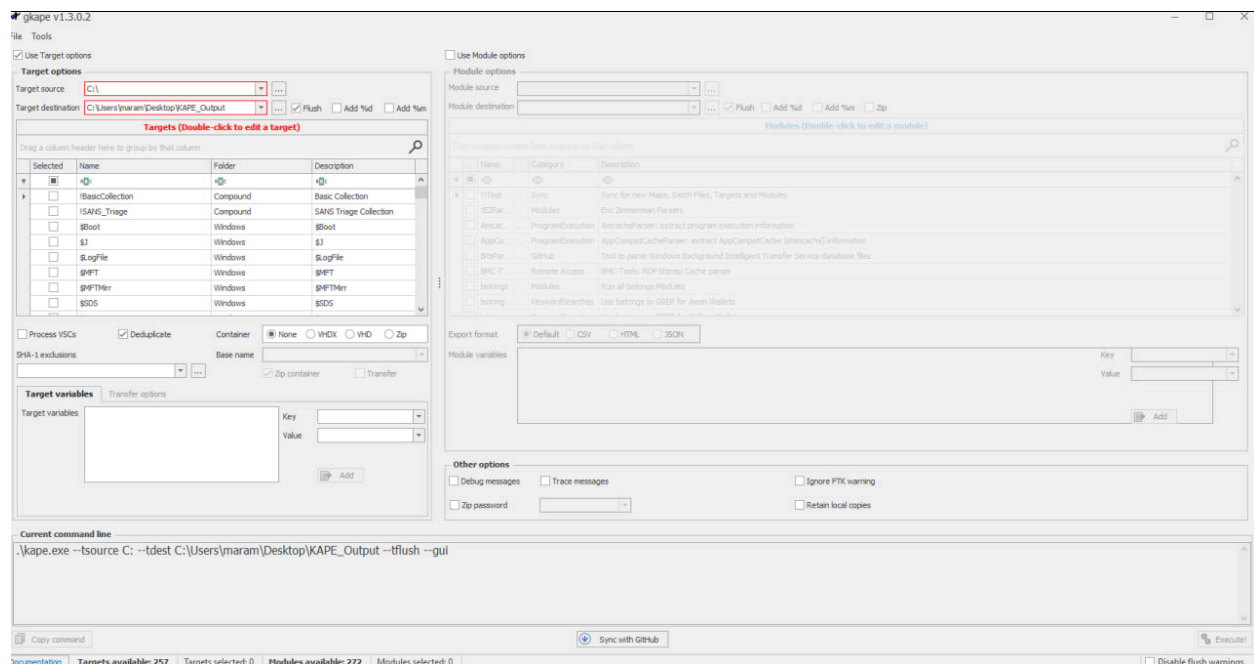
- **KAPE (Kroll Artifact Parser and Extractor)** – for triage collection
- **Registry Explorer** – for offline registry analysis
- **PECmd** – for Prefetch artifact analysis

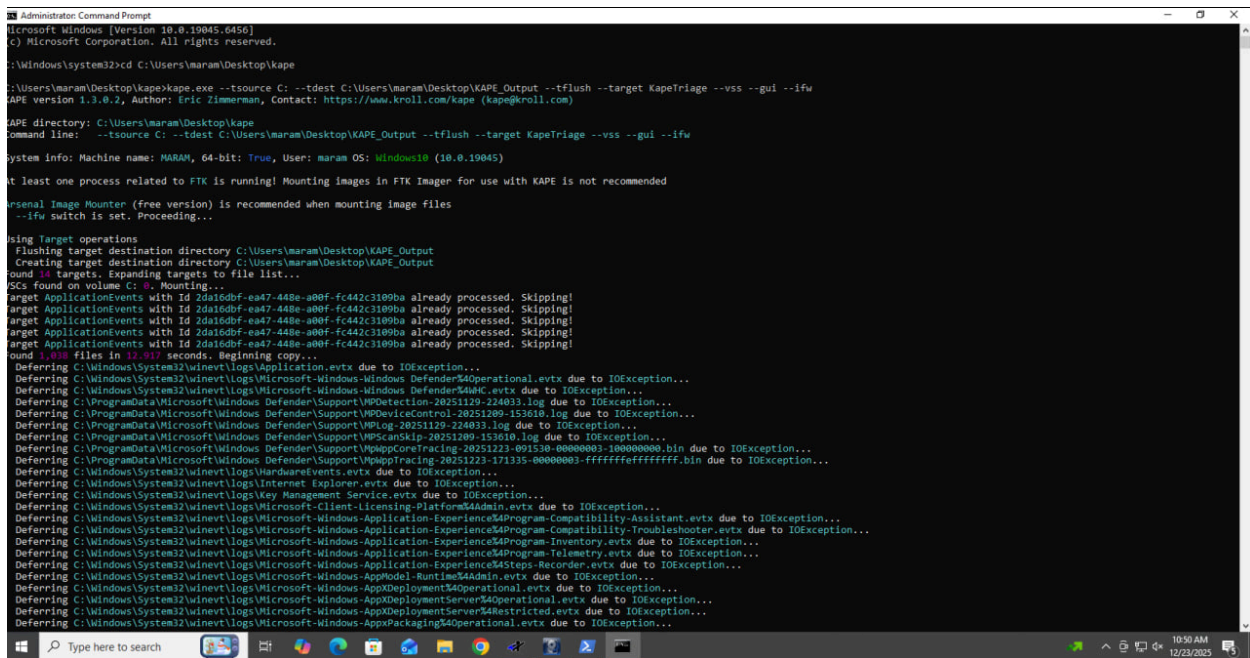
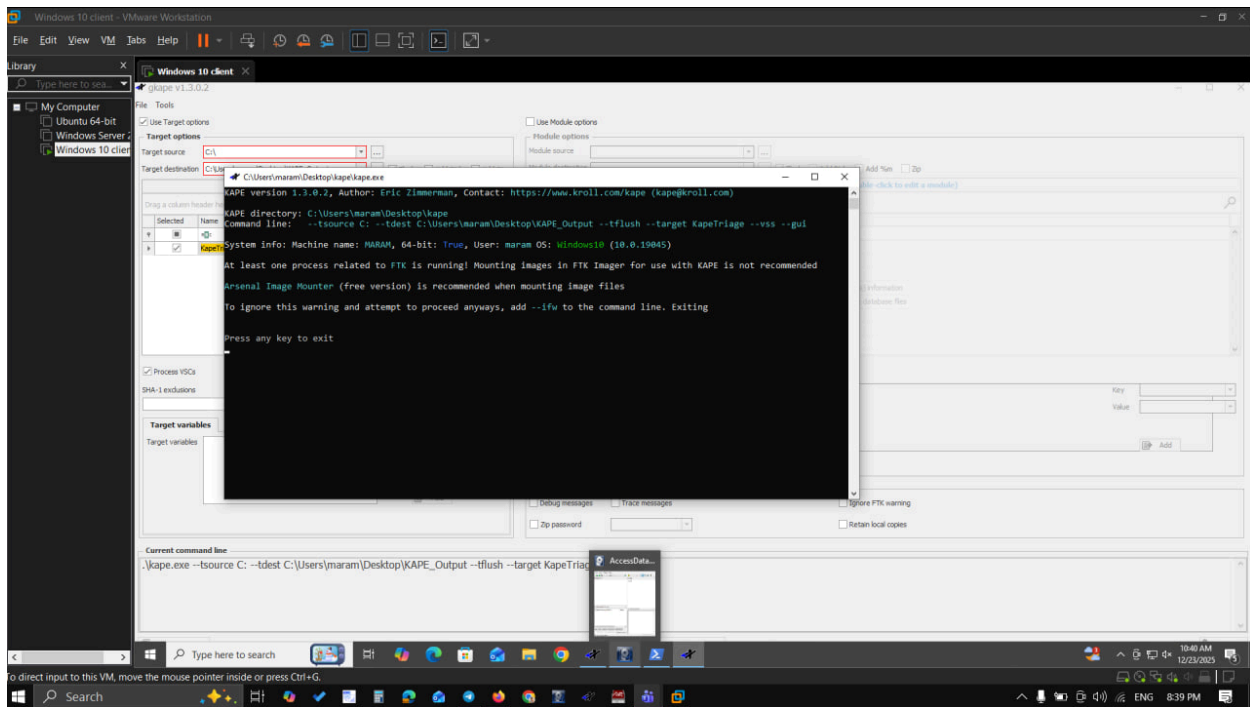
- **Event Viewer / Windows PowerShell.evtx** – for PowerShell and execution activity
- **Manual File System Analysis** – for user profiles and persistence validation

1) TRIAGE PACKAGE CREATION

- KAPE configuration
- Source and destination paths
- Selected targets/modules
- Successful triage execution







2) MACHINE INFORMATION (SYSTEM & SOFTWARE HIVES)

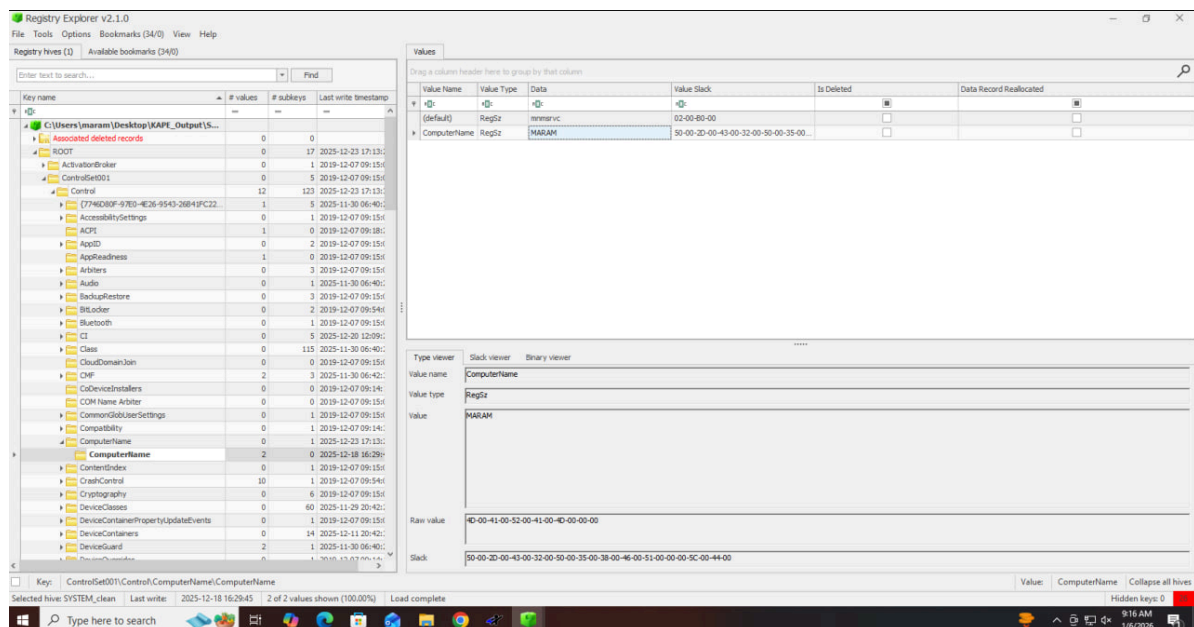
Findings

1. Machine Name

The computer name was identified from the **SYSTEM** registry hive.

Registry Path: **HKLM\SYSTEM\CONTROLSET001\COMPUTERNAME\COMPUTERNAME**

- **Value Data:** **MARAM**

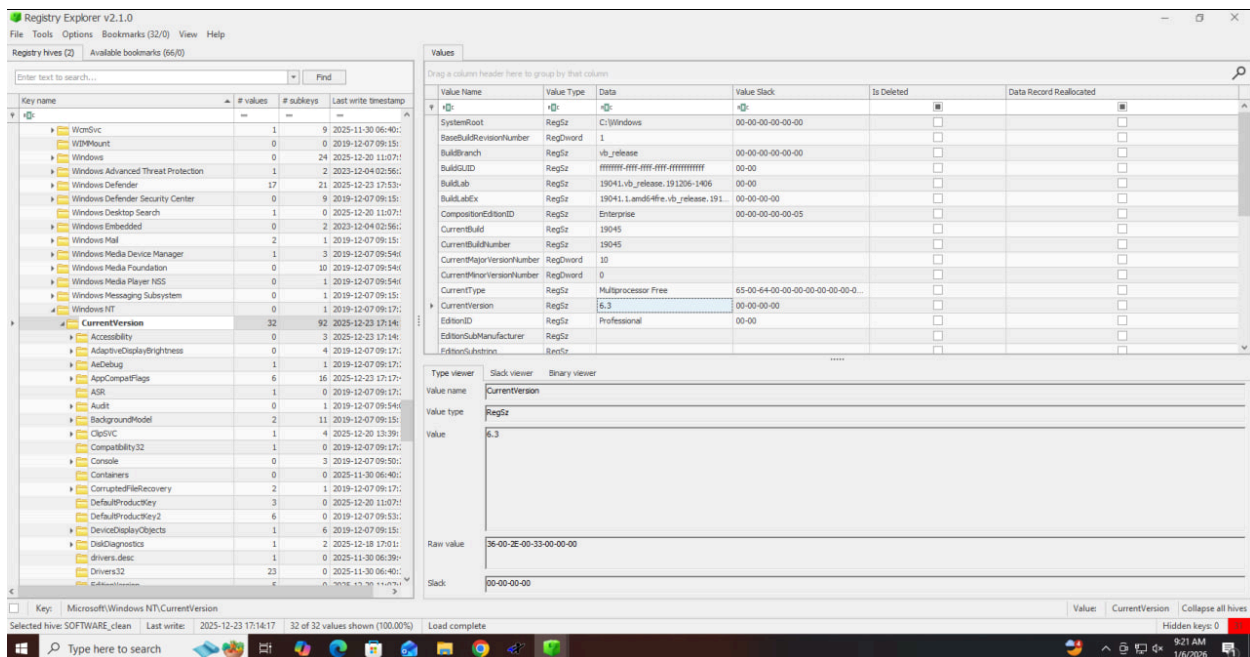


2. Windows Version

Windows version information was extracted from the SOFTWARE registry hive.

Registry Path: **HKLM\SOFTWARE\Microsoft\WindowsNT\CURRENT VERSION**

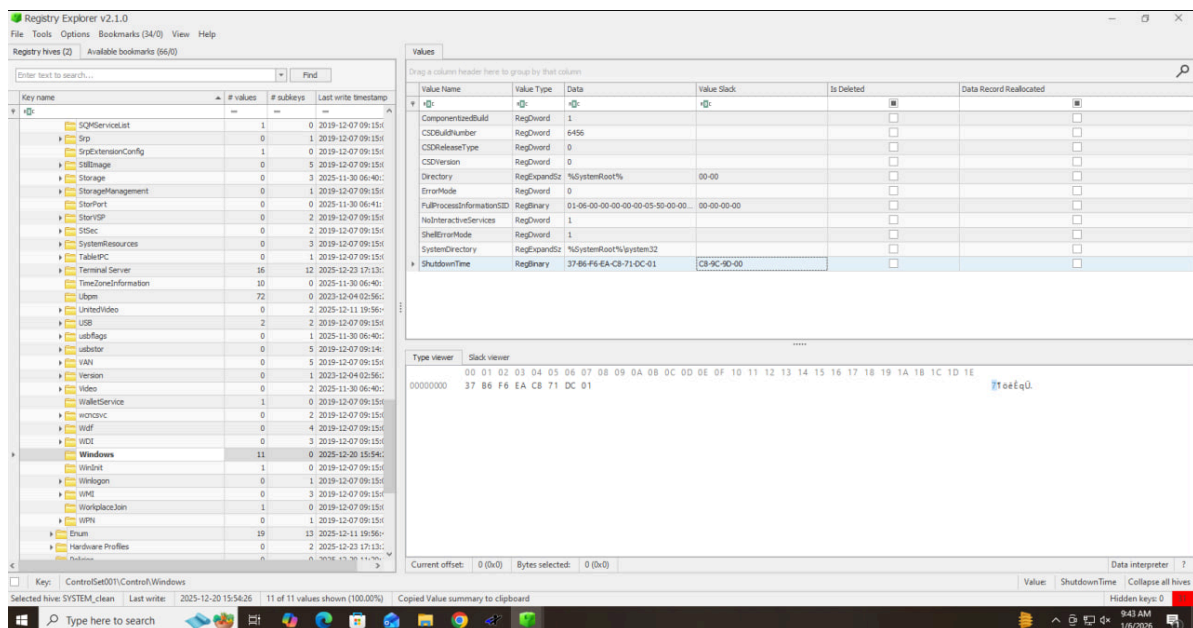
- **Value Data:** 6.3



3. Last Shutdown Time

The last system shutdown information was identified from the SYSTEM registry hive.

- **Registry Path:**
- **HKLM\SYSTEM\ControlSet001\Control\Windows**



3) SAM HIVE ANALYSIS

Analyze the SAM hive to identify local accounts and detect newly created or suspicious accounts.

Artifacts Examined

- SAM hive
- Users directory on disk

The SAM hive was examined to enumerate local users. Account creation times, group memberships, and profile existence were validated to identify potentially malicious accounts.

Extracted Account Details

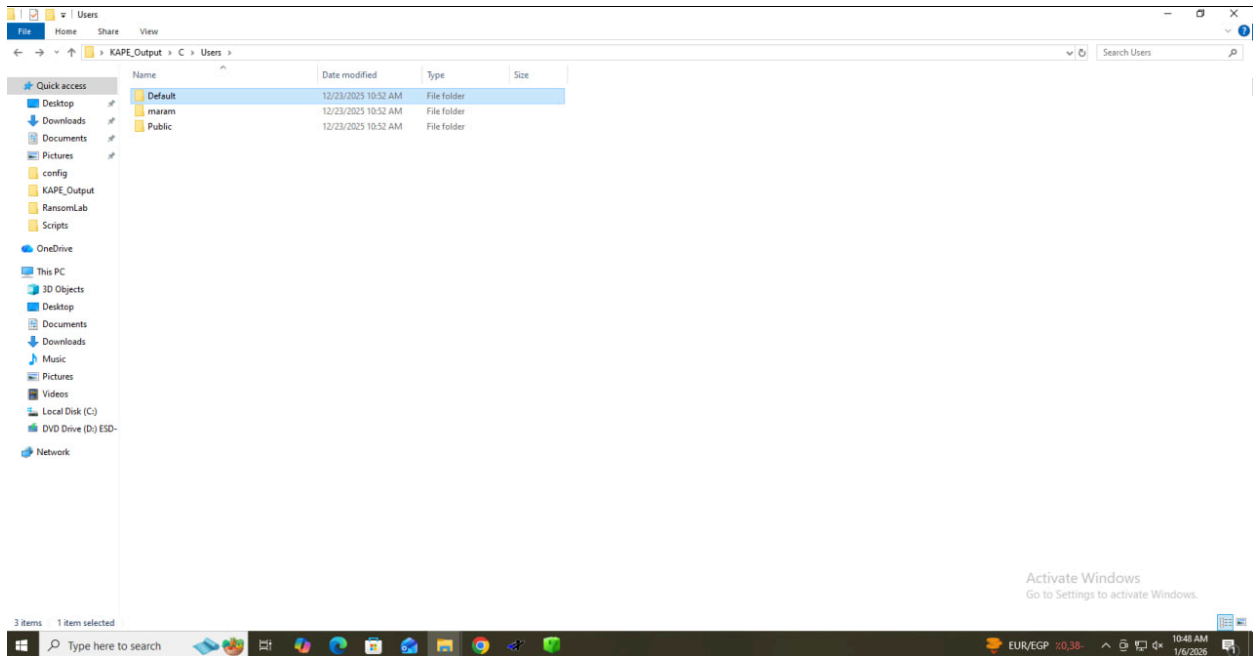
- Account Name
- Account Creation Time
- Group Membership

Registry Path: **SAM\SYSTEM\Domains\Accounts\Users**

The screenshot displays the Registry Explorer v2.1.0 interface. The left pane shows the tree structure of the registry, with the path **C:\Users\maram\Desktop\KAPE_Output\Values\Users** selected. The right pane shows a table of user accounts with the following columns: Valid, User, Inval, Total, Crea, Last, Last, Expr, Us, Full, Pass, Groups, Com, User, Hom, Inter, Rese, Acco, Hom, Pass, Tem, Norm, Mne, Inter, Work, Serv, Pass, Auto. The table contains 6 rows of data, including users like Administrator, Guest, and maram. The bottom pane shows the binary viewer for the selected value, displaying the raw value as 00-00-00-00.

Valid	User	Inval	Total	Crea	Last	Last	Expr	Us	Full	Pass	Groups	Com	User	Hom	Inter	Rese	Acco	Hom	Pass	Tem	Norm	Mne	Inter	Work	Serv	Pass	Auto
1	Administrator	0	0	2025-12-23 17:47:21	2025-12-23 17:47:21	2025-12-23 17:47:21		Administrators			Administrators, Users																
1	Guest	0	0	2025-12-23 17:47:21	2025-12-23 17:47:21	2025-12-23 17:47:21		Guests			Guests																
1	maram	0	0	2025-12-23 17:47:21	2025-12-23 17:47:21	2025-12-23 17:47:21		maram			Administrators																

- **User Profile Presence (C:\Users)**



The account had been created but was not used in the attack , so it does not have profile because it was not logged in with.

4) NTUSER.DAT ANALYSIS

Identify user actions that led to execution of the attack script via NTUSER.DAT hive analysis.

Artifacts Examined

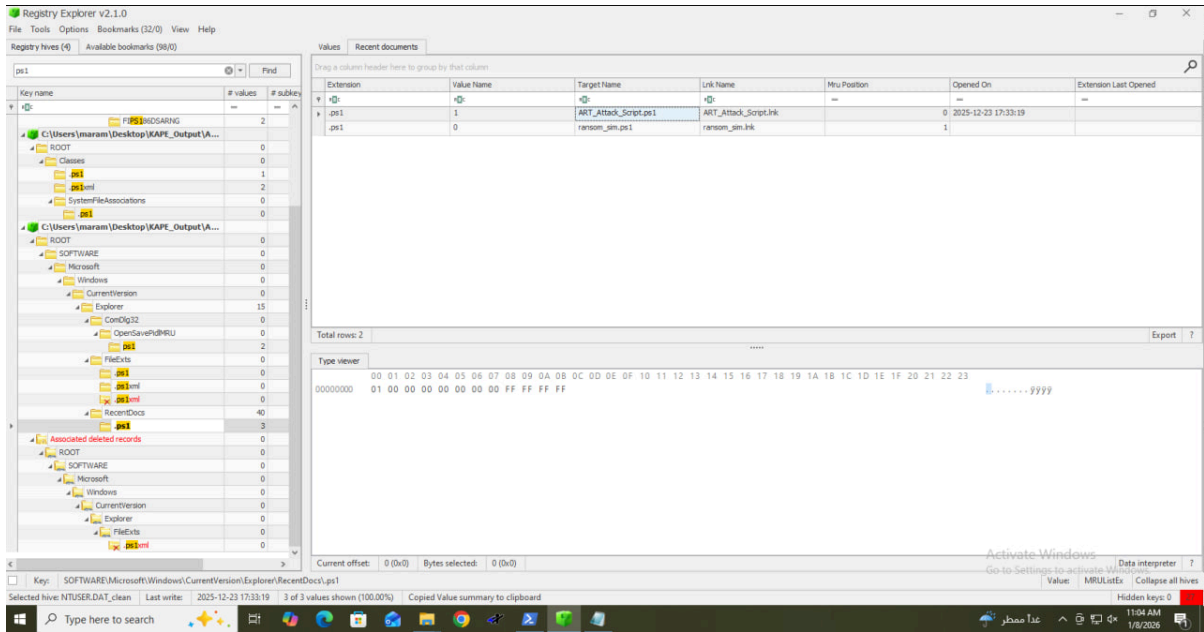
- NTUSER.DAT registry hive

The NTUSER.DAT hive was analyzed to reconstruct user actions preceding attack execution. Registry artifacts related to script execution and user interactions were correlated to establish the execution chain.

Identified Evidence

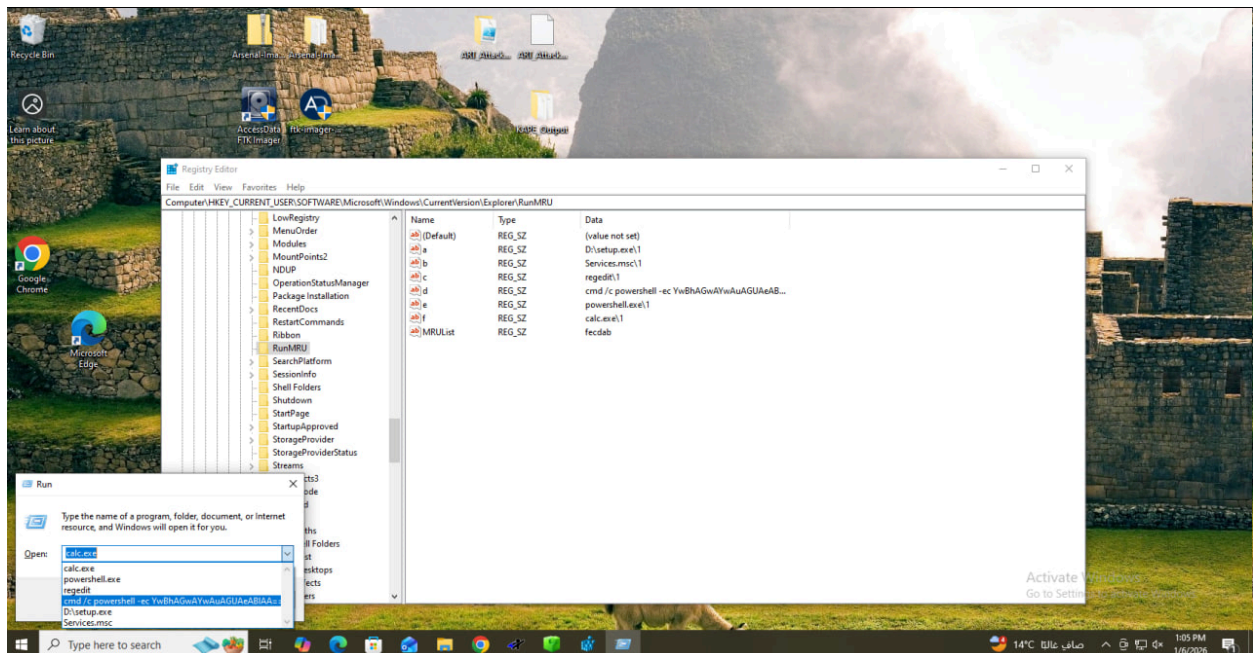
- **Registry path associated with executed script**

Registry Path: `SOFTWARE\Microsoft\Windows\CurrentVersion\Explorer\RecentDocs\ .ps1`



- Executable path that launched the script

C:\Users\maram\Desktop\KAPE_Output\SOFTWARE_clean



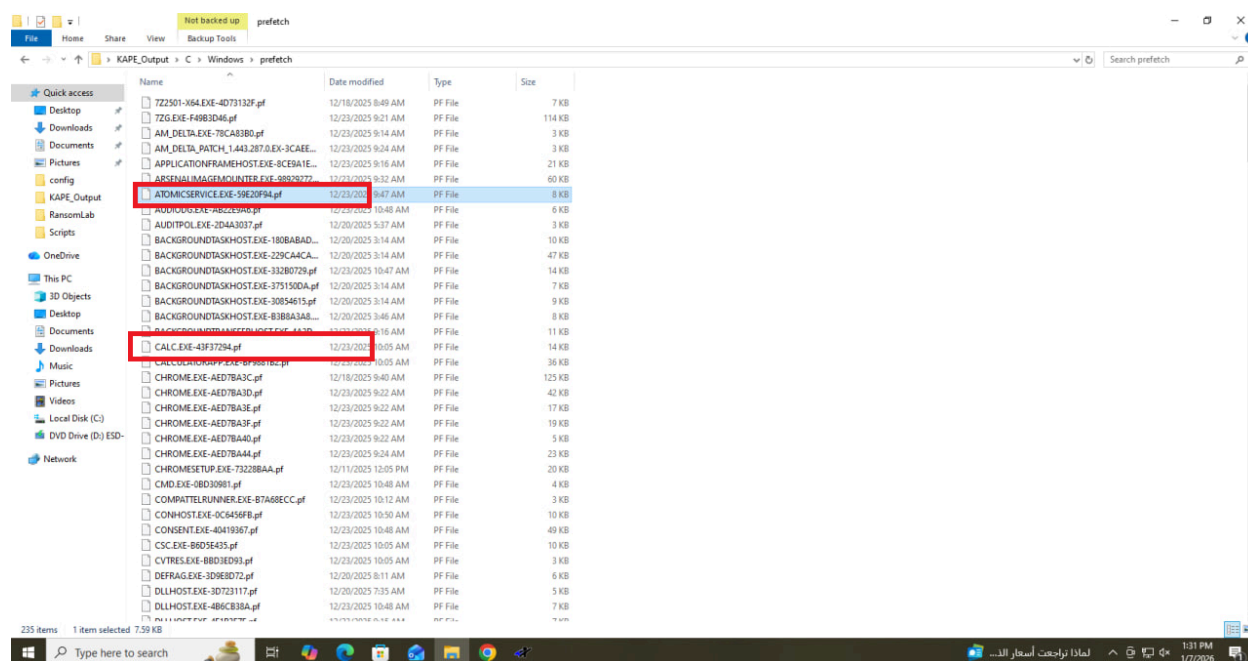
5) Execution Chain Analysis

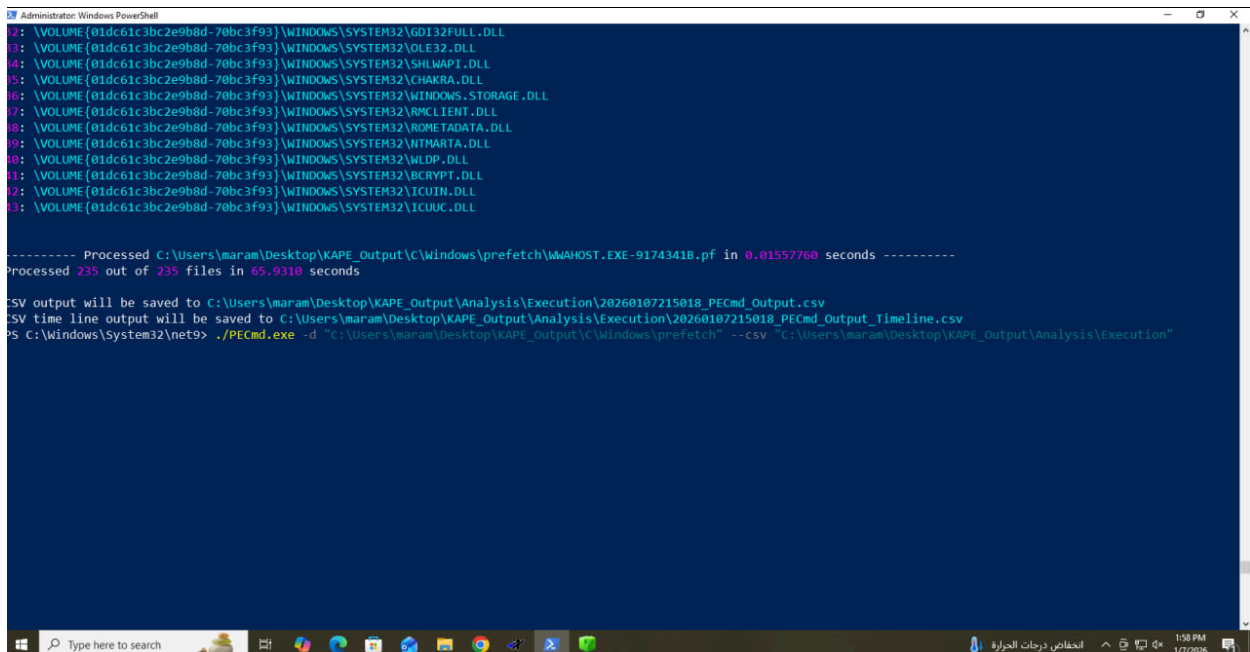
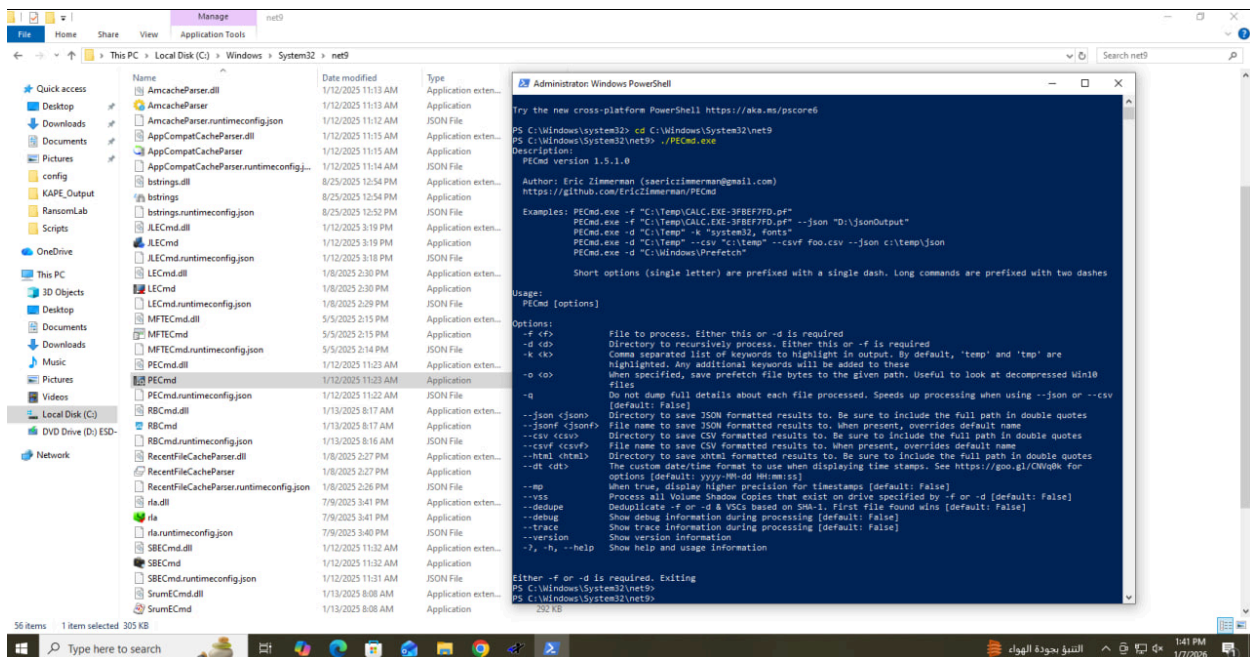
Identify execution behavior observed during the attack, including evidence of PowerShell usage, calculator execution, registry interactions, and service activity.

Artifacts Examined

- Prefetch files (PECmd)
- Registry-related process artifacts
- Service execution artifacts

Execution artifacts were analyzed to confirm behaviors during the attack. PowerShell activity was correlated with registry modifications and secondary process execution, including Calculator. Service-related artifacts were analyzed for links to Atomic Red Team activity





```
Administrator: Windows PowerShell
PS C:\Users\user\Desktop> .\Kape_Output\Kape\Windows\prefetch\MMUCLT.EXE-50573F8E.pf In 0.8385528 seconds -----
Processing C:\Users\user\Desktop\Kape_Output\Kape\Windows\prefetch\MMUCLT.EXE-50573F8E.pf
Created on: 2025-11-20 20:42:06
Modified on: 2025-11-20 20:42:06
Last accessed on: 2025-11-20 21:50:18
Executable name: MMUCLT.EXE
Hash: 91743418
File size (bytes): 25,232
Version: Windows 11
Run count: 1
Last run: 2025-11-20 20:42:06
Volume Information:
00: Name: \VOLUME{81d6c130c2e98d0-780c3f93} Serial: 780C3F93 Created: 2025-11-30 06:37:04 Directories: 9 File references: 58
Directories referenced: 9
01: \VOLUME{81d6c130c2e98d0-780c3f93}\PROGRAMDATA
02: \VOLUME{81d6c130c2e98d0-780c3f93}\PROGRAMDATA\MICROSOFT
03: \VOLUME{81d6c130c2e98d0-780c3f93}\PROGRAMDATA\MICROSOFT\WINDOWS
04: \VOLUME{81d6c130c2e98d0-780c3f93}\PROGRAMDATA\MICROSOFT\WINDOWS\APPREPOSITORY
05: \VOLUME{81d6c130c2e98d0-780c3f93}\PROGRAMDATA\MICROSOFT\WINDOWS\APPREPOSITORY\PACKAGES
06: \VOLUME{81d6c130c2e98d0-780c3f93}\PROGRAMDATA\MICROSOFT\WINDOWS\APPREPOSITORY\PACKAGES\MICROSOFT_WINDOWS_CLOUDEXPERIENCEHOST_10_0_19041_3636_NEUTRAL_NEUTRAL_CVRN1H2XYEV
07: \VOLUME{81d6c130c2e98d0-780c3f93}\WINDOWS
08: \VOLUME{81d6c130c2e98d0-780c3f93}\WINDOWS\SYSTEM32
Files referenced: 44
09: \VOLUME{81d6c130c2e98d0-780c3f93}\WINDOWS\SYSTEM32\WTL.DLL
10: \VOLUME{81d6c130c2e98d0-780c3f93}\WINDOWS\SYSTEM32\MMUCLT.EXE (Executable: True)
11: \VOLUME{81d6c130c2e98d0-780c3f93}\WINDOWS\SYSTEM32\KERNEL32.DLL
12: \VOLUME{81d6c130c2e98d0-780c3f93}\WINDOWS\SYSTEM32\USER32.DLL
13: \VOLUME{81d6c130c2e98d0-780c3f93}\WINDOWS\SYSTEM32\GDI32.DLL
14: \VOLUME{81d6c130c2e98d0-780c3f93}\WINDOWS\SYSTEM32\LOCALIZED
15: \VOLUME{81d6c130c2e98d0-780c3f93}\PROGRAMDATA\MICROSOFT\WINDOWS\APPREPOSITORY\PACKAGES\MICROSOFT_WINDOWS_CLOUDEXPERIENCEHOST_10_0_19041_3636_NEUTRAL_NEUTRAL_CVRN1H2XYEV\S-1-5-21-2961229852-1014687687-302527888-1000_PCKSDP
16: \VOLUME{81d6c130c2e98d0-780c3f93}\WINDOWS\SYSTEM32\APPHELP.DLL
17: \VOLUME{81d6c130c2e98d0-780c3f93}\WINDOWS\SYSTEM32\APPHELP.DLL
18: \VOLUME{81d6c130c2e98d0-780c3f93}\WINDOWS\SYSTEM32\APPHELP.DLL
19: \VOLUME{81d6c130c2e98d0-780c3f93}\WINDOWS\SYSTEM32\COMBASE.DLL
20: \VOLUME{81d6c130c2e98d0-780c3f93}\WINDOWS\SYSTEM32\COMBASE.DLL
21: \VOLUME{81d6c130c2e98d0-780c3f93}\WINDOWS\SYSTEM32\RPCRT4.DLL
22: \VOLUME{81d6c130c2e98d0-780c3f93}\WINDOWS\SYSTEM32\SHCORE.DLL
```

Registry Explorer v2.1.0

File Tools Options Bookmarks (0/0) View Help

Registry Hives (2) Available bookmarks (0/0)

Enter text to search...

Find

Key name # values # subkeys Last write

InventoryApplicationFile 2 438 2025-11-20 18:40:16

3dviewer.exe\123c2b... 19 0 2025-11-20 18:40:16

7z.exe\af6683e0fa326... 19 0 2025-11-20 18:40:16

7z2501-v64.exe\9d5579... 19 0 2025-11-20 18:40:16

7za.exe\20a374b59c7... 19 0 2025-11-20 18:40:16

7zip.exe\36a28795081... 19 0 2025-11-20 18:40:16

7zip.exe\36a28795081... 19 0 2025-11-20 18:40:16

adobe_gpt_gui.exe\de5... 19 0 2025-11-20 18:40:16

adobe.exe\76b19f6600... 19 0 2025-11-20 18:40:16

aggregatorhost.e\2f153... 20 0 2025-11-20 18:40:16

am_delta.exe\3025d818... 19 0 2025-11-20 18:40:16

am_delta_patch_1\4b1... 19 0 2025-11-20 18:40:16

apdhostregisr\12305... 19 0 2025-11-20 18:40:16

apostrophe.exe\0f47a1... 19 0 2025-11-20 18:40:16

apostrophepyth\50455... 19 0 2025-11-20 18:40:16

applicationname\7c5b... 20 0 2025-11-20 18:40:16

arsenalagemoun\8d8c... 19 0 2025-11-20 18:40:16

audodg.exe\6e948042... 20 0 2025-11-20 18:40:16

audtpol.exe\ef93e2597... 20 0 2025-11-20 18:40:16

authenticatorma\5a6c7... 19 0 2025-11-20 18:40:16

autochk.exe\9a0a7ac1... 20 0 2025-11-20 18:40:16

alc.exe\bc55004a5a7... 20 0 2025-11-20 18:40:16

colorchecker...8f18b... 19 0 2025-11-20 18:40:16

Values AmacheInventoryApplicationFile

Drag a column header here to group by that column

Timestamp	Path	Name	Product Name	Publisher	Version	SHA1
2025-12-20 13:24:03	c:\windows\system32\cmd.exe	cmd.exe	microsoft windows operating system	microsoft corporation	10.0.19041.4355 (winbuild.16010.10800)	d79c86f68139c8b914b458e509f879c72566c
2025-12-20 11:19:40	c:\programdata\microsoft\windows defender\platform\4.18.25110.6-0\mpcmdrun.exe	MpCmdRun.exe	microsoft windows operating system	microsoft corporation	4.18.25110.6 (7d793183-384c779116f6a6d3c3be997251d7)	ab381384499c6f8a52c2e62692178e79c54c6c
2025-12-09 15:24:44	c:\program files\windows defender\platform\4.18.25110.900-0\mpcmdrun.exe	MpCmdRun.exe	microsoft windows operating system	microsoft corporation	4.18.1909.6 (winbuild.16010.10800)	e2b4c5eefca33416f9f39a67a7240c345c5c1d
2025-12-18 15:50:28	c:\programdata\microsoft\windows defender\platform\4.18.25110.900-0\mpcmdrun.exe	MpCmdRun.exe	microsoft windows operating system	microsoft corporation	4.18.25100.900 (0a0fb177efae3a7a64896131727074bdc1379)	4e7ee13a6e84c66e65850e09e1c0b5d5d8c1
2025-12-18 15:50:29	c:\program files\vmware\tools\vmtoolsd\vmtoolsd.exe	VMwareTools.exe	vmware tools	broadcom inc.	12.5.3.52671	acab0ac73046b189b8db56e23d0fe8b5070f19
2025-12-18 15:50:29	c:\program files\vmware\tools\vmtoolsd\vmtoolsd.exe	VMwareTools.exe	vmware tools	broadcom inc.	12.5.3.52671	6c19b13114b2530c3f46c0f9b6c29464b9332

Total rows: 428

Type viewer Binary viewer

Value name WritePermissionsCheck

Value type RegDword

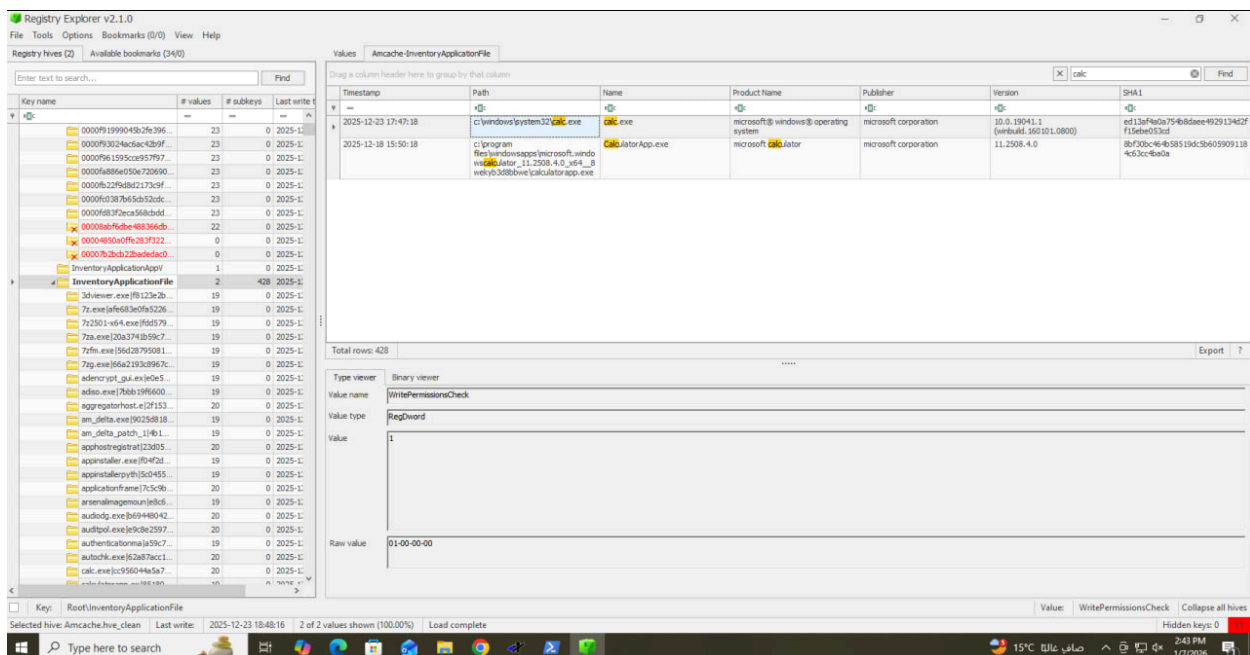
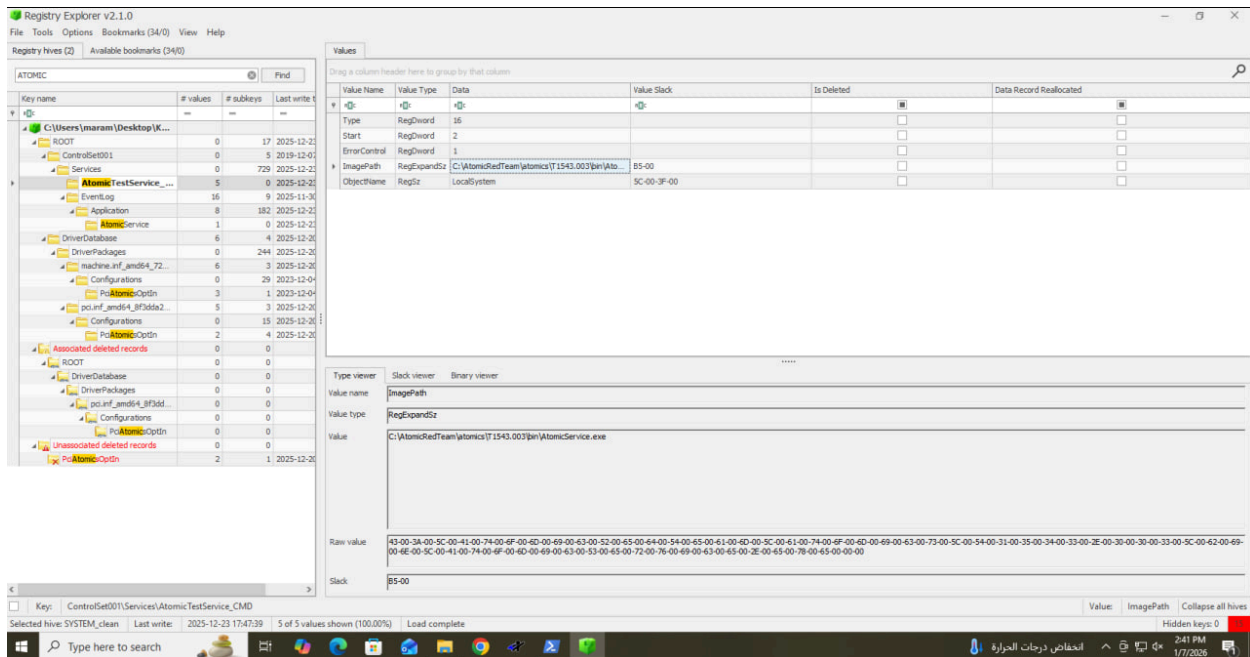
Value 1

Raw value 01-00-00-00

Key: Root\InventoryApplicationFile

Selected hive: AmacheInventoryApplicationFile Last write: 2025-12-23 18:40:16 2 of 2 values shown (100.00%) Load complete

Hidden keys: 0



Based on the analyzed artifacts, the execution chain can be summarized as follows:

1. PowerShell was executed on the system.
2. Atomic Red Team PowerShell scripts were launched.
3. A child process (Calculator) was spawned during script execution.

This confirms that PowerShell served as the primary execution vector, leading to successful process execution on the system.

6) Persistence Mechanism

Persistence Mechanism #1 – User-Level Registry Run Key

A persistence mechanism was identified within the current user's registry hive (**NTUSER.DAT**). The attacker created a registry value under the *Run* key, causing a malicious executable to be launched automatically each time the user logs in.

Artifact Details

- **Registry Hive:** NTUSER.DAT
- **Registry Path:**

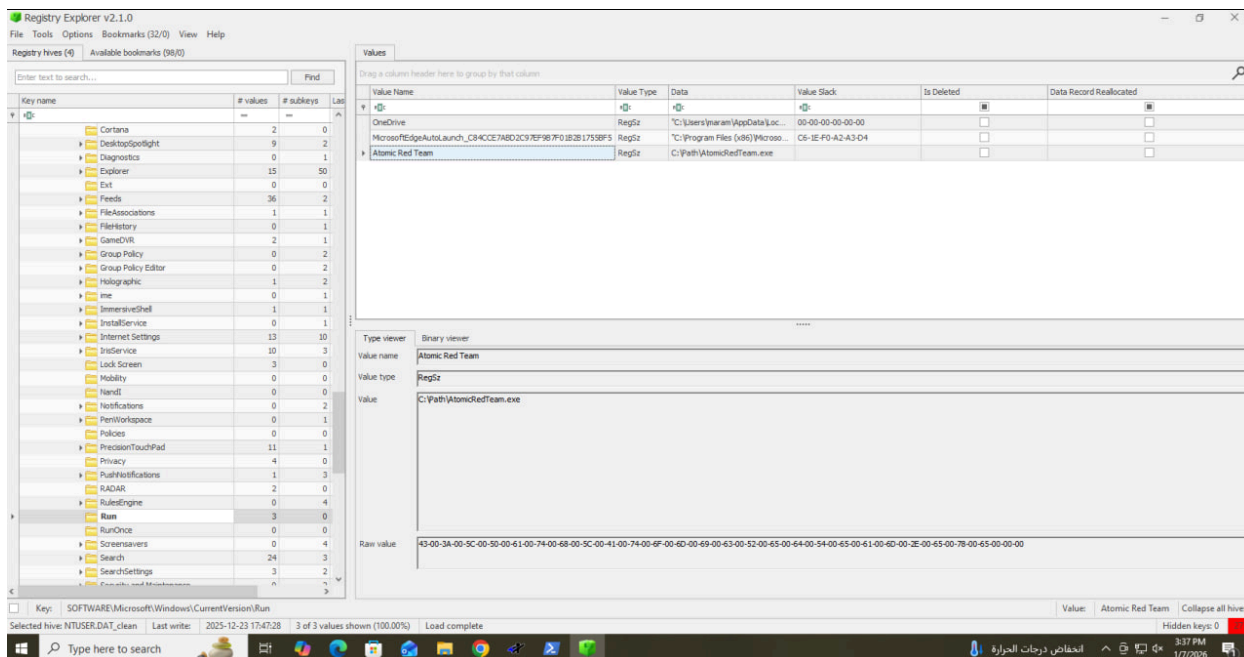
```
SOFTWARE\Microsoft\Windows\CurrentVersion\Run
```

- **Value Name:** Atomic Red Team
- **Value Type:** REG_SZ
- **Data:**

```
C:\Path\AtomicRedTeam.exe
```

- **Last Write Time:** 2025-12-23 17:47:28

This confirms the presence of **user-level persistence**, ensuring execution of the malicious payload upon user logon.



Persistence Mechanism #2 – Malicious Windows Service

A second persistence mechanism was identified in the **SYSTEM registry hive**, where a malicious Windows service was created. Services configured with an automatic start mode execute during system startup, allowing persistence even before any user logs in.

Artifact Details

- **Registry Hive:** SYSTEM
- **Registry Path:**

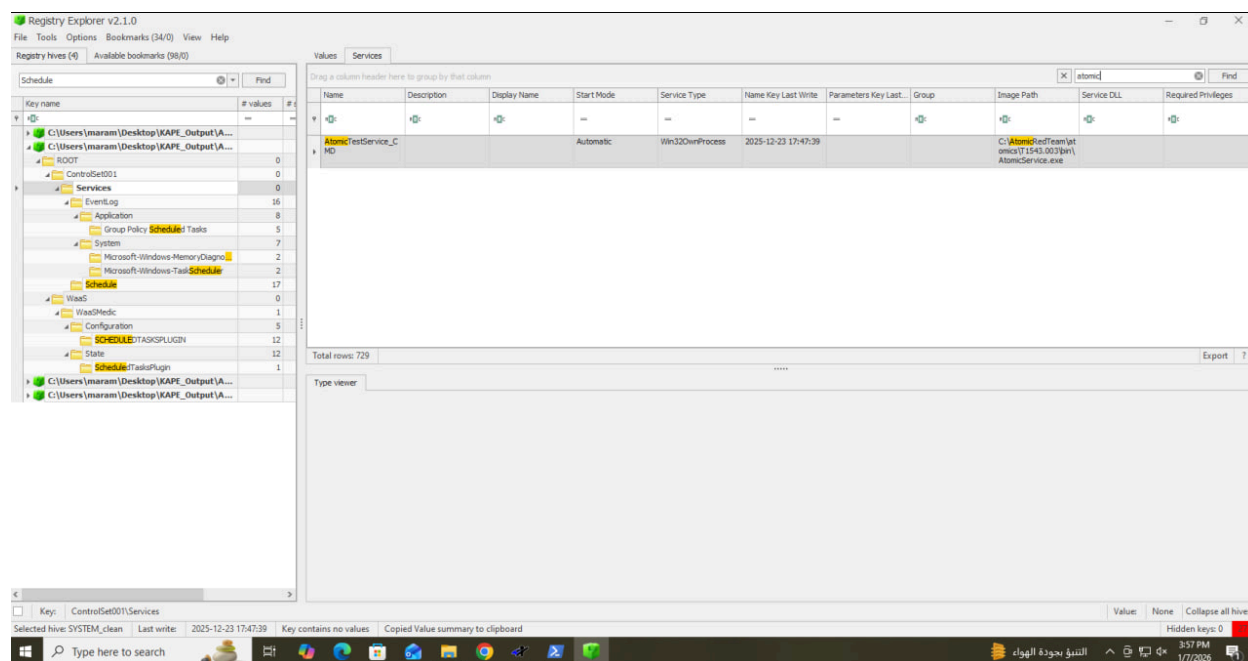
ControlSet001\Services\AtomicTestService_C

- **Service Name:** AtomicTestService_C
- **Start Mode:** Automatic
- **Service Type:** Win32OwnProcess
- **Image Path:**

C:\AtomicRedTeam\atomics\T1543.003\bin\AtomicService.exe

- **Last Write Time:** 2025-12-23 17:47:39

This confirms **system-level persistence** via a malicious service, ensuring execution on every system boot.



Remediation Actions

- Removal of malicious registry entries
- Deletion of associated executables or scripts
- Validation of system startup and scheduled tasks
- Verification that no unauthorized services remain active

MITRE ATT&CK Technique Execution Results

T1078.003 – Valid Accounts: Local Account Creation

T1059.001 – Command and Scripting Interpreter: PowerShell (Fileless Execution)

T1547.001 – Boot or Logon Autostart Execution: Registry Run Key

T1053.005 – Scheduled Task / Job: Scheduled Task on Logon

T1543.003 – Create or Modify System Process: Windows Service

Conclusion

The investigation successfully reconstructed the attack timeline, identified the execution chain, and uncovered multiple persistence mechanisms. Using systematic registry and artifact analysis, the compromised system was remediated and restored to a secure state.